

2.3 Population & the Environment

1. Population Change

Population change: The increase or decrease in the number of people in an area over time. Determined by the balance between birth rate, death rate, and net migration.

Natural change = Birth rate – Death rate

Population change = (Births – Deaths) + (Immigration – Emigration)

Global population reached 8 billion in November 2022 and is projected to peak at ~10.3 billion in the mid-2080s before gradually declining (UN 2024 revision).

Key Demographic Indicators

Indicator	Definition	Global Average (approx. 2024)
Crude Birth Rate (CBR)	Number of live births per 1,000 population per year	~18 per 1,000
Crude Death Rate (CDR)	Number of deaths per 1,000 population per year	~8 per 1,000
Rate of Natural Increase (RNI)	CBR – CDR, expressed as a percentage	~1.0%
Total Fertility Rate (TFR)	Average number of children born per woman during her lifetime	~2.3 (halved from 4.7 in 1970)
Infant Mortality Rate (IMR)	Number of deaths of infants under 1 year per 1,000 live births	~28 per 1,000 (ranges from 2 in Japan to 80+ in Sierra Leone)

Life Expectancy	Average number of years a person is expected to live from birth	~73 years (ranges from 54 in Chad to 85 in Japan)
Replacement Level Fertility	TFR of ~2.1 needed to maintain stable population without migration	— (over half of all countries are now below replacement)
Dependency Ratio	$(\text{Population } 0-14 + 65+) / \text{Population } 15-64 \times 100$	Varies: Niger ~110 (young dependents), Japan ~70 (elderly dependents)

Factors Affecting Birth Rates

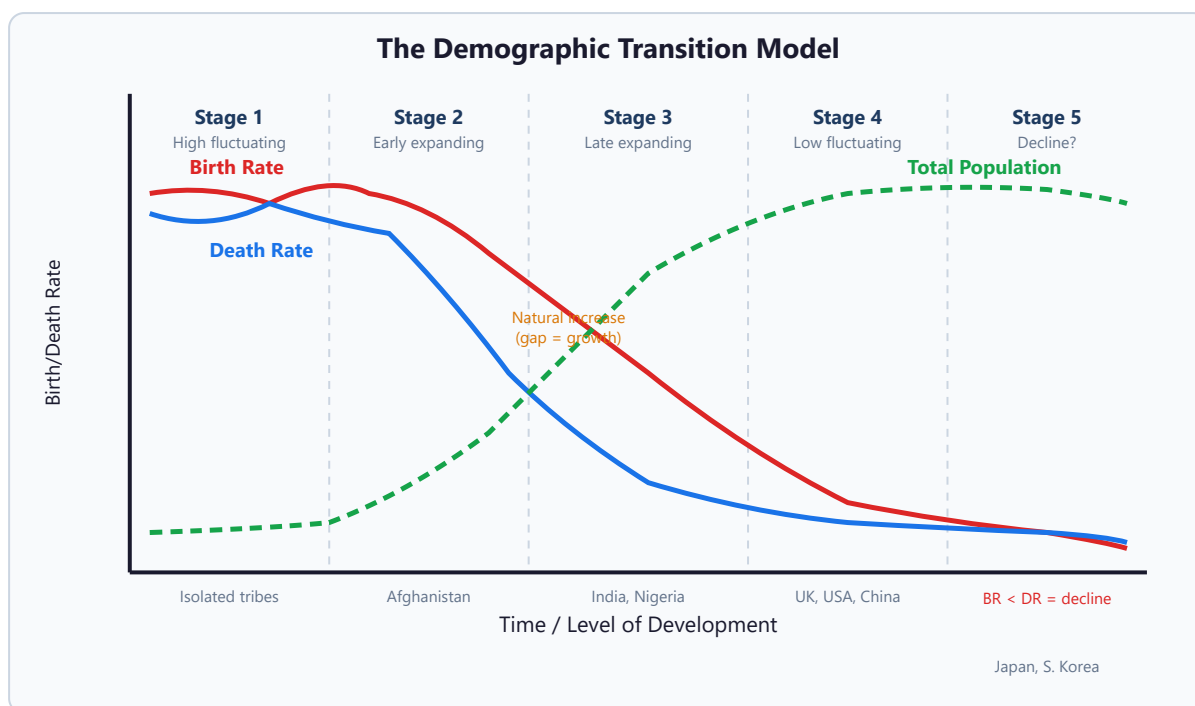
- **Education (especially female)** — the single strongest predictor of fertility decline. Women with secondary education have on average 1.5 fewer children than those with no education. Education delays marriage, increases contraceptive use, and empowers women with career alternatives.
- **Access to contraception** — availability and cultural acceptance of family planning. In Sub-Saharan Africa, 24% of women who want to avoid pregnancy lack access to modern contraception (unmet need).
- **Economic development** — as countries develop, children shift from economic assets (farm labour) to economic costs (education, childcare). The "cost of childhood" rises sharply with urbanisation and modernisation.
- **Cultural and religious factors** — attitudes to family size, roles of women, religious teachings on contraception (e.g. Catholic Church, some Islamic interpretations). However, cultural factors interact with economic conditions — Catholic Italy and Spain have among the world's lowest TFRs.
- **Government policies** — pro-natalist (France: generous paid parental leave, childcare subsidies, tax breaks for larger families; Sweden: 480 days shared parental leave) or anti-natalist (China's former one-child policy, India's forced sterilisation programme 1975–77).
- **Urbanisation** — urban living reduces birth rates: smaller housing, higher cost of children, less need for farm labour, greater access to contraception and education, more career opportunities for women.
- **Infant mortality** — high IMR encourages larger families as "insurance" against child deaths. As IMR falls, parents gain confidence that children will survive, and fertility declines with a lag.
- **Status of women** — where women have greater legal rights, workforce participation, and decision-making power, fertility tends to be lower.

Factors Affecting Death Rates

- **Healthcare quality** — access to modern medicine, vaccinations (global childhood immunisation now covers ~85%), antibiotics, surgery, maternal care. Universal healthcare systems dramatically reduce preventable deaths.
- **Nutrition and food security** — adequate calorie and micronutrient intake. The Green Revolution (1960s–80s) doubled food production and prevented widespread famine.
- **Clean water and sanitation** — reduces waterborne diseases (cholera, typhoid, dysentery). 2 billion people still lack safely managed drinking water (2023).
- **Disease prevalence** — HIV/AIDS devastated life expectancy in Southern Africa (Botswana's LE fell from 65 to 49 between 1990 and 2002, now recovered to 62 with antiretroviral treatment). Malaria kills ~600,000/yr, mostly children in Sub-Saharan Africa.
- **Conflict and instability** — war, civil unrest, displacement raise death rates directly and by destroying healthcare, food systems, and infrastructure.
- **Age structure** — countries with ageing populations (Japan, Germany, Italy) have higher crude death rates even with excellent healthcare, because a large proportion of the population is elderly.
- **Lifestyle factors** — in HICs, non-communicable diseases (heart disease, cancer, diabetes, obesity-related conditions) are now the leading causes of death. Smoking, alcohol, diet, and sedentary lifestyles.

2. The Demographic Transition Model (DTM)

Demographic Transition Model: A theoretical model describing how population change progresses through stages as a country develops economically. It shows the relationship between birth rates, death rates, and total population over time. Originally based on the experience of Western European countries during industrialisation.



DTM Stage Details

Stage	BR	DR	Population	Characteristics	Example Countries
1: High fluctuating	High (~40)	High (~40)	Low, stable	Pre-industrial. No contraception, children needed for labour, high IMR. Disease, famine, poor sanitation cause high DR.	No modern countries; isolated Amazonian tribes
2: Early expanding	High (~40)	Falling (~20)	Rapidly growing	DR falls due to improved healthcare, sanitation, nutrition, and water. BR stays high (cultural lag). Large natural increase. "Population explosion."	Afghanistan (TFR 4.6), Mali, Chad, Niger
3: Late expanding	Falling (~25)	Low (~10)	Still growing but slowing	BR falls due to urbanisation, female education, contraception, changing role	India (TFR 2.0), Nigeria (TFR ~4.5–5), Egypt, Kenya

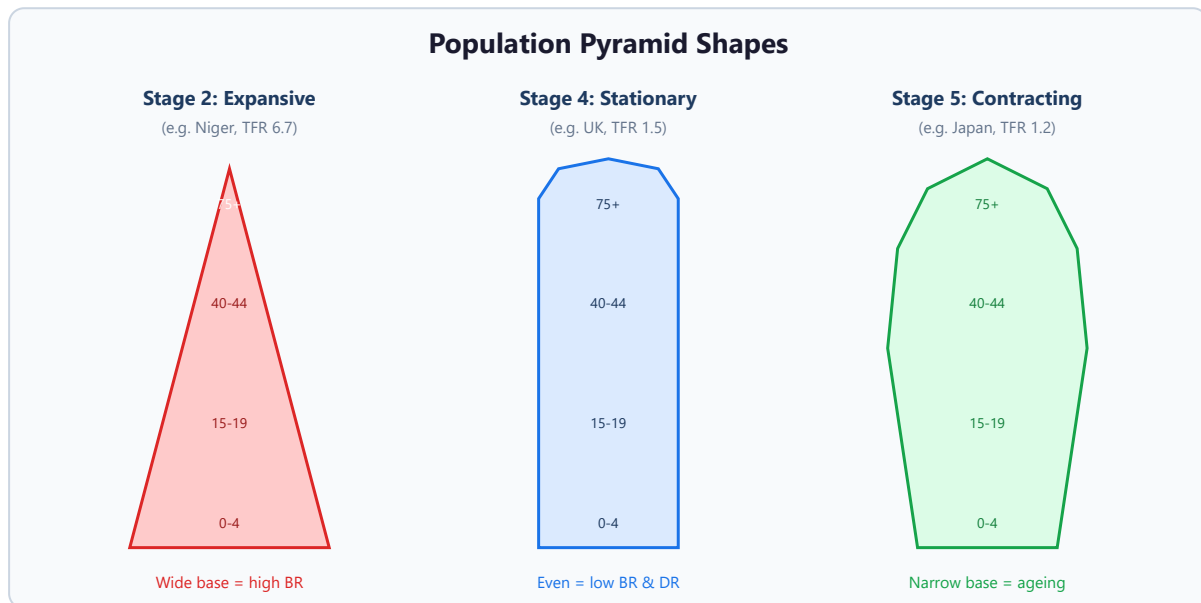
				of women, lower IMR. Natural increase narrows.	
4: Low fluctuating	Low (~10–12)	Low (~10–12)	High, stable	Both rates low. Population stable or growing slowly through immigration. Ageing population structure. Most HICs.	UK (TFR ~1.4), USA (1.6), France (1.6), China (1.0)
5: Decline (debated)	Very low (<10)	Exceeds BR	Declining	BR falls well below replacement. Rapid ageing, rising dependency ratio, labour shortages, pension crises. Population shrinks without immigration.	Japan (TFR 1.2), South Korea (0.7), Italy (1.2), Germany (1.4)

Limitations of the DTM

- Based on the Western European historical experience — may not apply to countries with different cultures, economies, or political systems.
- Does not account for **migration** — a major component of population change in many countries (e.g. UAE, Singapore, UK).
- Assumes all countries follow the **same linear pathway** — ignores unique political/cultural contexts (e.g. HIV/AIDS reversed death rate decline in Southern Africa; China's one-child policy artificially accelerated the transition).
- Stage 5 is **contested** — not universally accepted. Some demographers argue it should include the possibility of fertility recovery.
- Does not predict **how long** each stage lasts — UK took 200 years; Thailand transitioned from Stage 2 to 4 in just 40 years.
- **Eurocentrism** — the model was developed to describe European industrialisation and may impose a Western trajectory on diverse development paths.
- Does not account for the **epidemiological transition** — the shift from infectious diseases to chronic/non-communicable diseases as the main causes of death.

3. Population Structure

Population pyramid: A graphical representation of the age-sex structure of a population. The shape reveals key characteristics: a wide base indicates a youthful population (high birth rates), a column shape indicates a stable population (low birth and death rates), and a top-heavy or inverted shape indicates an ageing/declining population.



Ageing Populations

Many HICs and some MICs face rapidly ageing populations (increasing proportion aged 65+). This results from sustained low birth rates and increasing life expectancy. Japan is the world's oldest country: 29% of the population is 65+, with a median age of 49.

Consequences of ageing:

- **Rising dependency ratio** — fewer working-age people supporting more elderly dependents. Japan's ratio is projected to reach 1 worker per 1 retiree by 2050 (currently 2:1).
- **Increased government spending** on pensions (Japan spends ~10% of GDP on pensions), healthcare (the elderly consume 3–5x more healthcare per capita), and social care (dementia care costs alone are projected to double by 2040 in the UK).
- **Labour shortages** — shrinking workforce potentially slowing economic growth. Germany's working-age population is projected to shrink by 7 million by 2035.

- **Changing demand for services** — more care homes, hospitals, and accessible infrastructure; fewer schools, maternity wards, and children's services.
- **Need for immigration** to fill workforce gaps — Germany admitted 1 million+ migrants in 2015 partly to address labour shortages.
- **"Grey vote"** — elderly voters become politically dominant, potentially skewing policy priorities towards pensions and healthcare rather than education and young families.

Youthful Populations

Many LICs have youthful populations (high proportion under 15). Niger has the world's youngest population: 50% are under 15, with a median age of just 15.4 years.

Consequences:

- **Enormous pressure on education** (building schools, training teachers), healthcare (maternal and child services), and employment (creating millions of jobs for young people entering the workforce).
- **"Youth bulge"** — a disproportionately large young adult cohort. If education and employment are inadequate, this can lead to social unrest, political instability, and migration (Arab Spring protests were partly driven by youth unemployment). But if harnessed, it creates a **demographic dividend**.
- **Demographic dividend** — if education and employment are provided for the large working-age cohort, the low dependency ratio can drive rapid economic growth. East Asian tigers (South Korea, Taiwan, Singapore) benefited from this in the 1970s–90s.
- **High dependency ratio** in the short term (many children to support relative to workers).
- **Momentum of growth** — even after fertility falls, population continues to grow for decades because the large young cohort is yet to have children.

4. Migration

Migration: The permanent or semi-permanent movement of people from one place to another. *International migration* crosses national borders; *internal migration* occurs within a country (e.g. rural-urban migration). There are an estimated 304 million international migrants globally (2024, UN DESA), representing 3.6% of the world's population.

Lee's Model of Migration (Push-Pull Theory, 1966)

Everett Lee proposed that migration decisions are influenced by four factors:

- **Push factors** at the origin — poverty, unemployment, conflict, persecution, natural disasters, environmental degradation, lack of services
- **Pull factors** at the destination — employment, higher wages, safety, education, healthcare, family reunification, freedom, quality of life
- **Intervening obstacles** — distance, cost of travel, border controls, immigration policies, language barriers, family ties, discrimination
- **Personal factors** — age (young adults most mobile), education, gender, risk tolerance, family situation, information networks

Types of Migration

Type	Description	Examples and Data
Economic migration	Voluntary movement to improve employment and income	Polish workers to UK post-2004 (~1M); Mexican workers to USA (~11M foreign-born Mexicans in US); Indian IT workers to Gulf States
Forced migration / Refugees	People forced to flee by conflict, persecution, or disaster	~123 million forcibly displaced worldwide (end of 2024, UNHCR). Largest refugee populations include Syrians, Afghans, Ukrainians and Venezuelans (roughly 6 million each)
Asylum seekers	People who have applied for refugee status but not yet received a decision	UK received asylum claims from ~84,000 people in 2023 and a record ~108,000 in 2024; Germany received 351,000 first-time applications in 2023
Internal (rural-urban)	Movement from rural areas to cities within the same country	China: 300M+ rural-urban migrants since 1980; India: 450M+ internal migrants; Nigeria: Lagos grows by ~600,000/yr
Retirement migration	Older people moving to areas with better climate or lifestyle	UK retirees to Spain, France, Portugal; UK internal migration to Devon, Cornwall, Dorset

Counter-urbanisation	City to rural/small town movement	Accelerated post-COVID: remote workers leaving London for rural/coastal areas. UK saw a rise in net moves from cities to rural and coastal areas in 2020–21
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Impacts of Migration

	Source Country	Destination Country
Positive	Remittances boost economy (global remittances to LMICs: \$656 billion in 2022, 3x total overseas aid), reduced pressure on resources/jobs, return migrants bring skills and capital, diaspora networks facilitate trade	Fills labour shortages (especially healthcare, agriculture, construction), cultural diversity and innovation, tax revenue, addresses ageing populations, entrepreneurship (40% of Fortune 500 companies were founded by immigrants or their children)
Negative	Brain drain (loss of skilled workers — 1 in 9 African-trained doctors works in an OECD country), ageing population left behind, family separation, dependency on remittances, loss of working-age population	Pressure on housing/services in areas of high concentration, social tensions and anti-immigrant sentiment, wage depression in low-skill sectors (debated), cultural integration challenges, exploitation of undocumented workers

Case Study: EU Migration to the UK (2004–2020)

Following EU enlargement in 2004 (A8 countries including Poland, Lithuania, Latvia joined), the UK experienced significant immigration. Between 2004 and 2016, an estimated 1 million Polish nationals moved to the UK, making Polish the second most common language. **Positive impacts:** filled labour shortages in agriculture (picking and packing), construction, hospitality, healthcare (~5.5% of NHS staff in England in 2019 were EU nationals), and food processing; net fiscal contributors (paid more in taxes than received in benefits — UCL study found A8 migrants contributed 12% more in taxes than they received in public services); cultural diversity, entrepreneurship (Polish shops, restaurants, churches). **Negative impacts:** pressure on housing and public services in some areas (e.g. Boston, Lincolnshire — which voted 76% Leave in the EU referendum), concerns about wage depression in low-skill sectors (disputed by most economists), contributed to the Brexit vote (2016) as immigration was a key campaign issue. **Post-Brexit:** EU net migration fell from 189,000/yr (2015) to near zero (2021), causing acute labour shortages in agriculture (crops left unharvested), hospitality (high

vacancy rates), social care (165,000 vacancies), and HGV driving. Non-EU migration subsequently increased to compensate.

Case Study: Syrian Refugee Crisis (2011–present)

The Syrian Civil War (from 2011) created the largest refugee crisis since World War II. By 2024: over 6 million Syrian refugees had fled, mainly to neighbouring countries (Turkey ~3 million, Lebanon 800,000+ registered, Jordan ~620,000) and Germany (~1 million Syrians), and ~7 million were internally displaced. After the fall of the Assad government in December 2024, over 1.5 million refugees and around 2 million IDPs had returned home by mid-2026 (UNHCR), though conditions remain fragile. **Impacts on host countries:** Turkey spent \$40+ billion hosting refugees; Lebanon's population increased by 25%, overwhelming infrastructure, schools, and healthcare; Jordan's water-scarce resources were strained further. Germany admitted 890,000 asylum seekers in 2015 alone ("Willkommenskultur" policy of Angela Merkel), sparking a political backlash (rise of AfD party, debate about integration). **Impacts on Syria:** devastating brain drain (doctors, engineers, teachers fled), destroyed human capital, generation of children missing education. **Broader significance:** the crisis exposed the limitations of the international refugee system, contributed to anti-immigration politics across Europe, and demonstrated how conflict creates cascading humanitarian, economic, and political consequences across multiple countries.

5. Population-Environment Relationships

Key Theories

Malthus (1798) — An Essay on the Principle of Population: Population grows geometrically (1, 2, 4, 8, 16...) while food production grows arithmetically (1, 2, 3, 4, 5...). This means population will always tend to outstrip food supply, leading to "positive checks" (famine, disease, war) that reduce population back to the level the land can support. Malthus also identified "preventive checks" (moral restraint, delayed marriage) that could voluntarily limit growth. A fundamentally pessimistic/neo-Malthusian view. Neo-Malthusians (Paul Ehrlich's "The Population Bomb" 1968, Club of Rome's "Limits to Growth" 1972) updated this argument for the modern era, warning of ecological collapse.

Boserup (1965) — The Conditions of Agricultural Growth: Population growth is not a threat but a stimulus for agricultural innovation. When population pressure increases, people develop new techniques to produce more food from the same land (e.g. irrigation, fertilisers, crop rotation, selective breeding, terracing, double-cropping). "Necessity is the mother of invention." An optimistic view that has been partially supported by the Green Revolution, but questioned by concerns about environmental sustainability.

Club of Rome / Limits to Growth (1972): Computer modelling (by Dennis Meadows et al.) predicted that continued exponential population and economic growth would lead to resource depletion and environmental collapse by the mid-21st century if growth trajectories were not changed. A neo-Malthusian perspective. The models have been updated (2004, 2012) and broadly track real-world trends, though critics argue technology can shift resource limits. The concept of "planetary boundaries" (Rockstrom et al., 2009) identifies 9 Earth system thresholds, of which 6 had been crossed by 2023 and 7 by the 2025 assessment.

Simon (1981) — The Ultimate Resource: Julian Simon argued that human ingenuity is the "ultimate resource." More people means more ideas, innovation, and problem-solving capacity. Population growth drives economic growth and technological progress. He won a famous bet with Paul Ehrlich (1980) that commodity prices would fall over a decade despite population growth (they did). A cornucopian/techno-optimist view.

Ecological Footprint and Carrying Capacity

Ecological footprint: The area of biologically productive land and water required to produce the resources consumed and absorb the waste generated by a given population. Measured in global hectares (gha) per person. The USA's ecological footprint is ~8.1 gha/person; India's is ~1.2 gha/person; the global average is ~2.8 gha/person. Earth's total biocapacity is only ~1.6 gha/person, meaning humanity is in ecological overshoot — using resources 1.75x faster than Earth can regenerate them. "Earth Overshoot Day" (when humanity has used a year's worth of resources) fell on 1 August in 2024 and 24 July in 2025.

Carrying capacity: The maximum population size that an environment can sustain indefinitely, given the available resources. It is not fixed — technology can increase it (Green Revolution, renewable energy) while environmental degradation can reduce it (soil erosion, water depletion, climate change). For Earth as a whole, estimates range from 2 billion (at US consumption levels) to 15 billion (at subsistence levels) — carrying capacity depends on lifestyle choices as much as absolute resource limits.

6. Global Population Futures and Population Policies

The UN projects global population will reach ~9.7 billion by 2050 and peak at ~10.3 billion in the mid-2080s before gradually declining (2024 revision). Almost all growth will occur in Sub-Saharan Africa (projected to grow from ~1.2 billion to over 2 billion by 2050) and South Asia. Meanwhile, populations in Europe, East Asia, and parts of Latin America are already declining or will soon.

The world is experiencing a dramatic **fertility decline**: global TFR fell from 4.7 (1970) to 2.3 (2024). Over half of all countries are now below replacement fertility (2.1). South Korea's TFR of 0.7 (2023) is the lowest ever recorded in any country. This creates a "demographic divergence" — some countries face youth bulges and rapid growth, while others face ageing and decline.

Case Study: China's Population Policies

China's **one-child policy (1979–2015)** was the world's most ambitious anti-natalist programme. Methods: financial penalties for additional children (fines of 3–10x annual income), rewards for compliance (priority housing, healthcare, education), workplace monitoring, and in some areas forced sterilisations and abortions. **Results:** TFR fell from 5.9 (1970) to 1.6 (2000). The government claimed the policy prevented ~400 million births (a figure many demographers dispute). **Unintended consequences:** severe gender imbalance (sex-selective abortions led to 120 boys per 100 girls at peak, creating a "bare branches" problem — 30–40 million men unable to find partners), rapidly ageing population (65+ population grew from 5% in 1980 to 15% in 2024, projected 30% by 2050), "4-2-1" family structure (one child supporting 2 parents and 4 grandparents), "little emperor" syndrome, rural enforcement abuses and human rights violations. **Policy reversal:** two-child policy (2016), three-child policy (2021), with penalties for extra births scrapped the same year, now actively encouraging births through cash incentives (a national childcare subsidy of 3,600 yuan (~US\$500) a year per child under 3 from 2025, plus local payments), extended maternity leave (158+ days), childcare subsidies, and housing discounts. **Despite all this, TFR continues to fall (~1.0 in 2023)** as

high living costs, housing prices, education expenses, changing attitudes toward marriage, and rising female workforce participation keep families small. China's population peaked in 2021 (~1.41 billion, official data) and has fallen every year since 2022 — projected to fall to ~630 million by 2100 (UN 2024). China demonstrates that once fertility decline becomes culturally embedded, it is extremely difficult to reverse.

Case Study: Uganda — Rapid Population Growth

Uganda (population ~46 million, 2024 census) has one of the world's fastest growth rates (~3.2%/yr) and highest TFRs (~4.7). It is transitioning from DTM Stage 2 to early Stage 3: death rates have fallen significantly (under-5 mortality dropped from 175 to 43 per 1,000 between 1990 and 2023) but birth rates remain high. **Reasons for persistent high fertility:** limited contraceptive access (only 35% of married women use modern contraception; 28% unmet need), cultural preference for large families (children as security, status, and farm labour), low female secondary education enrollment (41%), 75% rural population, early marriage (34% of girls married before 18), and limited government investment in family planning. **Consequences:** 47% of population is under 15, creating enormous pressure on education (classroom sizes of 100+ common), healthcare (1 doctor per 10,000 people vs WHO recommendation of 1 per 1,000), employment (900,000 young people enter the job market annually, but only 75,000 formal jobs are created), and the environment (deforestation rate of 2.7%/yr for farmland expansion, soil erosion, Lake Victoria pollution). **Potential:** if education expands (especially for girls) and contraceptive access improves, Uganda could capture a demographic dividend from its large young workforce. But without these investments, the youth bulge risks becoming a source of instability.

Case Study: Japan — Population Decline and Super-ageing

Japan is the world's most extreme example of population ageing and decline. TFR is 1.2 (well below replacement of 2.1). Population peaked at 128.1 million (2008) and is now 123 million (2024), projected to fall to 87 million by 2070 and to ~50 million by 2120 (national projections). 29.1% of the population is 65+ (the highest proportion globally). **Causes:** high cost of living and child-rearing in cities, long working hours (incompatible with family life), high female education and career aspirations, limited childcare, cultural shift away from marriage (30% of Japanese aged 30–34 have never married), small housing. **Consequences:** ~850,000 more deaths than births in 2023 (over 900,000 in 2024), severe labour shortages (especially in healthcare, construction, agriculture — 1.5 million unfilled jobs), rising pension and healthcare costs (social security spending: 24% of GDP), abandoned rural towns ("ghost villages" — 9 million vacant homes), shrinking domestic market and

tax base. **Government response:** robotics and AI to replace workers, gradual opening to immigration (foreign workers doubled to 2 million between 2015 and 2023, but Japan remains culturally resistant to large-scale immigration), childcare expansion (free preschool from 2019), financial incentives (some municipalities offer \$7,000+ per child), but TFR has continued to fall. Japan's Prime Minister declared in 2023 that addressing the fertility crisis was "now or never" — a challenge that may prove impossible to solve through policy alone once cultural norms have shifted.

Exam Tip: AQA expects you to evaluate different perspectives on population-environment relationships. Malthusian and Boserupian views represent opposite ends of a spectrum. The reality is nuanced: the Green Revolution supports Boserup (technology increased food production), but environmental limits are real and growing (climate change, water scarcity, soil degradation support neo-Malthusian concerns). Use specific evidence to build a balanced argument. The best answers acknowledge that the relationship depends on consumption patterns and technology, not just population numbers.

Exam Tip: Always consider the social, ethical, and political dimensions of population policies. China's one-child policy achieved demographic goals but at enormous human cost. Pro-natalist policies (France, Sweden) have limited success because fertility decisions are deeply personal. Discuss whether governments should intervene in family size, and whose rights take priority. This ethical dimension elevates answers beyond mere description.

Exam Tip: When discussing population change, always distinguish between population SIZE and population GROWTH RATE. Global population is still growing in absolute terms but the growth rate has halved since the 1960s. Similarly, distinguish between the PERCENTAGE of elderly and the ABSOLUTE NUMBER. Japan has a high percentage of elderly but a smaller population than Nigeria, which has a much larger absolute number of young people. Precision with data impresses examiners.

7. Pro-natalist Policies in Depth

France: Europe's Most Successful Pro-natalist State

Case Study: France — Pro-natalist Policy

France has maintained a relatively high TFR (1.8–2.0 for much of the 2000s and 2010s) compared to other Western European countries (EU average ~1.5) through sustained, comprehensive pro-natalist policies developed over more than a century. French pro-natalism has roots in the demographic shock of the Franco-Prussian War (1870–71) and World War I, when France suffered disproportionate population losses compared to Germany.

Key policies:

Family allowances (Allocations familiales): Monthly cash payments to all families with two or more children, regardless of income. In 2024, approximately €150/month for two children and €340/month for three. No means testing for the basic allowance (though some supplements are income-related). This is explicitly natalist — payments increase with each additional child.

Parental leave: 16 weeks' maternity leave at full pay (100% salary replacement) rising to 26 weeks for a third child. The PreParE scheme provides extended leave (up to 3 years per child) with an allowance, particularly used by lower-income parents. Shared parental leave encouraged though take-up by fathers remains low (~2%).

Childcare: *Crèches* (state-funded nurseries) provide highly subsidised care from birth to 3 years, with fees capped on an income-sliding scale. *Écoles maternelles* (state preschools) provide free, universal childcare/education from age 3. This addresses the key barrier to fertility in most HICs: the incompatibility of work and childcare for women.

Tax system: The *quotient familial* system reduces income tax liability for larger families. Each child reduces the household's taxable income fraction, creating an increasing financial advantage for three or more children. Estimated value: ~€5,000 per year tax saving for a family with three children.

Housing: Priority access to social housing for families with children; specific large-family housing allocations.

Outcomes: France's TFR (1.68 in 2023) remains among the highest in the EU, although it fell to ~1.6 in 2024. Population continues to grow slowly (unlike most EU countries), maintaining a relatively balanced age structure. Annual cost to the French state: approximately €50–60 billion (>2% of GDP on family policy).

Limitations: France's TFR has declined from 2.0 (2010) to 1.68 (2023) despite sustained policy commitment, suggesting that structural economic factors (housing costs, youth unemployment,

rising cost of living) and cultural shifts (delayed marriage, career prioritisation) are overwhelming policy support. The policy cannot reverse cultural change once embedded.

Singapore: From Anti-natalist to Pro-natalist

Case Study: Singapore — Policy Reversal

Singapore's population policy is one of the world's most dramatic reversals. From 1966 (when the Family Planning and Population Board was set up), Singapore ran an anti-natalist campaign (TFR was ~4.7), from 1972 using the slogan **"Stop at Two"**: financial penalties for third and subsequent children (lower priority for school places; higher hospital fees for additional births); paid sterilisation; and active promotion of small family norms. The policy succeeded: TFR fell from 4.7 (1966) to 2.1 (1975) and below replacement by 1977.

By 1984, TFR had fallen to 1.6 and Singapore faced rapid ageing and labour shortages. In 1987 the government reversed course: the **"Have Three, or More If You Can Afford It"** slogan replaced "Stop at Two." But the anti-natalist cultural norms proved difficult to reverse.

Current pro-natalist package (2024): The Baby Bonus scheme provides SGD 11,000 per child (first/second) and SGD 13,000 for third+ children (as a combination of cash gift and government co-savings in a Child Development Account). 16 weeks' paid maternity leave (plus new shared parental leave from 2025). Government-subsidised childcare (up to 80% of costs). Priority housing allocation for young families buying their first flat within 4 km of parents (Proximity Housing Grant). Fertility treatment subsidised up to 75% at government hospitals (up to 3 IVF cycles).

Result: Despite comprehensive, expensive policy support, Singapore's TFR was 0.97 in 2023 — one of the lowest in the world. Singapore's experience demonstrates that once urbanisation, high costs, and gender-egalitarian norms take hold, fertility is essentially impossible to return to replacement level through policy alone. The government acknowledges it relies on immigration to maintain population, with Singapore admitting ~20,000–25,000 new citizens and ~30,000–35,000 new permanent residents annually.

8. Anti-natalist Policies in Depth

China's One-Child Policy: A Detailed Analysis

China's one-child policy (1979–2015) is covered in detail in the earlier case study, but additional dimensions are worth highlighting for AQA:

Dimension	Detail
Implementation variation	The policy was not uniformly applied. Ethnic minorities (55 groups including Tibetans, Uyghurs, Mongols) were generally permitted 2–3 children. Rural families whose first child was a girl were often permitted a second child. Urban residents, especially government workers and party members, faced strict enforcement.
Gender imbalance legacy	Sex ratio at birth peaked at 121:100 (boys:girls) in 2008. An estimated 30–40 million Chinese men (the "bare branches") cannot find female partners. This has driven: trafficking of women from Vietnam and Myanmar; demand for commercial sex workers; delayed marriage and family formation; and complex social anxieties about masculinity and status.
"4-2-1 problem"	One child supporting two parents and four grandparents creates an unprecedented eldercare burden. China's current working-age (15–59) population is already shrinking by ~7 million/year. The pension system (originally designed for a different demographic structure) is projected to face insolvency by the 2030s without reform.
Human rights dimension	Documented abuses during enforcement: forced sterilisations, forced abortions (including late-term abortions), fines confiscating family assets, dismissal from government jobs, demolition of homes. Reports documented very large numbers of coerced sterilisations and abortions under the policy. These abuses were more common in rural areas and under local officials facing pressure to meet quotas.
Economic consequences	The "only child generation" (post-1980s) fuelled China's rapid economic growth: a large working-age population, low dependency ratio, and high savings rate drove GDP growth. This "demographic dividend" is now exhausted as the workforce shrinks and the elderly population grows rapidly.

Exam Tip: AQA questions on population policy require you to evaluate both **effectiveness** (did the policy achieve its demographic goals?) and **impact** (what were the intended and unintended social, economic, ethical, and environmental consequences?). China's one-child policy was demographically

effective but ethically damaging and created long-term structural problems. France's pro-natalist policy is ethically benign but limited in effectiveness. These contrasts make for rich evaluative arguments.

9. Malthus vs Boserup: Evaluation Table

Dimension	Malthus / Neo-Malthusian view	Boserup / Cornucopian view	Evidence and evaluation
Core argument	Population growth outstrips food supply, leading to famine, disease, war (positive checks) or moral restraint (preventive checks).	Population growth stimulates agricultural innovation; necessity drives invention; technology can always raise food production.	Both have partial support. Global food production per capita has increased (Boserup supported), but ~670 million (2024) remain hungry (distribution failure) and environmental limits are increasingly evident (Malthus partially vindicated).
Role of technology	Technology provides only temporary relief; population rebounds to consume any gains (Malthusian trap).	Technology is the primary response to scarcity; past crises (Irish Famine, Green Revolution) show human adaptability.	Green Revolution: doubled global cereal yields (1960s–80s), preventing widespread famine (Boserup). But based on fossil fuel inputs, aquifer depletion, and seed biodiversity loss — may not be sustainable (Neo-Malthusian caveat).
Environmental limits	Real and binding: land, water, and biodiversity have finite limits that growing populations must ultimately respect.	Limits are not fixed: technology expands carrying capacity; genetic engineering, desalination, vertical farming extend possibilities.	Planetary boundaries concept (Rockstrom, 2009) identifies 9 biophysical limits, of which 7 had been transgressed by 2025. Climate change, biodiversity loss, and nitrogen cycle disruption represent genuine Malthusian limits. But new technologies (precision

			fermentation, solar energy) may partially escape these constraints.
Distribution vs production	Even if production keeps pace, maldistribution means many still starve; focusing on production ignores inequality.	The problem is not just production but also access, governance, and market failure; growth can be redistributed through policy.	Global food production is theoretically sufficient to feed everyone, yet ~670 million (2024) are hungry. This suggests Boserup is right that production capability exists, but Malthus is also right that distribution fails to translate production into food security for all.
Timeframe	Short-term: immediate pressure on resources triggers checks. Long-term: environmental degradation reduces future carrying capacity.	Medium-to-long-term: innovation takes time but humanity has always found solutions before hitting absolute limits.	Timing matters: if environmental tipping points are crossed before innovation catches up, Malthusian collapse remains possible even if Boserupian technology eventually emerges.
Who is affected?	All populations, though poorest and most resource-dependent are most vulnerable.	Benefits of innovation are not evenly distributed; wealthier countries and individuals adopt new technology first.	Green Revolution benefited large farmers more than smallholders; GM crops are patented by corporations; desalination is affordable only to wealthy nations. Technology can be Boserupian in aggregate but Malthusian in its distributional effects.

Carrying Capacity: A Critical Examination

Carrying capacity debates: The concept of a fixed maximum population the Earth can support is controversial. Estimates range from 2 billion (Paul Ehrlich, at US consumption levels) to 10–15 billion (at subsistence) to effectively unlimited (Julian Simon, with technological optimism). The key insight is that carrying capacity is not a fixed number — it depends on **consumption patterns** (Americans use 8 times as many resources as Indians), **technology** (solar energy, lab-grown meat, GMO crops all

expand theoretical capacity), and **governance** (efficient distribution vs waste). The question "how many people can Earth support?" cannot be answered without specifying the lifestyle those people would live.

The **IPAT formula** (Paul Ehrlich and John Holdren, 1971) captures the multi-factor nature of environmental impact:

$$I = P \times A \times T$$

Where I = total environmental Impact, P = Population size, A = Affluence (consumption per person), T = Technology (environmental impact per unit of consumption). This shows that impact can be reduced by lowering affluence (consumption), improving technology (efficiency), or reducing population. A growing population with decreasing per-capita consumption (as in many LICs) may have less impact than a stable population with rapidly rising consumption (as in China or India's growing middle classes). The IPAT formula highlights that focusing solely on population numbers is misleading — consumption patterns are equally or more important.

10. Food Systems: The Green Revolution and Beyond

The Green Revolution (1960s–1980s)

Green Revolution: A package of agricultural technologies (high-yielding varieties of wheat, rice, and maize; synthetic fertilisers; pesticides; and irrigation) developed and disseminated from the 1960s onwards, primarily targeting South Asia, East Asia, and Latin America. Norman Borlaug, the leading plant scientist, received the 1970 Nobel Peace Prize for his work on high-yielding wheat varieties.

Positive Impacts	Negative Impacts
Global cereal production doubled between 1960 and 1985, outpacing population growth.	Required expensive inputs (fertilisers, pesticides, irrigation) that only wealthier farmers could afford, widening rural inequality.
India became self-sufficient in wheat production by 1974 (Punjab Wheat Revolution), avoiding	Intensified water use: Punjab (India) aquifer depleted at unsustainable rates; groundwater table falling 50 cm/yr

the famines predicted by neo-Malthusians. Food prices fell significantly.	in parts.
Crop yields per hectare increased 2–3x compared to traditional varieties. Mexico's wheat yields tripled; Philippine rice yields doubled.	Loss of crop genetic diversity: traditional varieties abandoned; 75% reduction in rice genetic diversity in the Philippines. Monocultures vulnerable to new pests/diseases.
Calorie production per capita in Asia increased ~50% between 1970 and 2000. Malnutrition rates fell significantly.	Environmental damage: pesticide contamination of waterways and soils; fertiliser runoff causing eutrophication; soil compaction and salinisation from intensive irrigation.
Freed up labour from subsistence farming, enabling industrialisation in South Korea, Taiwan, China.	Many small farmers were unable to compete with larger farms using Green Revolution technology and were displaced into cities (contributing to megacity growth).
Demonstrated that technology can increase carrying capacity (Boserup vindicated). Global famine did not occur despite rapid population growth.	Bypassed Sub-Saharan Africa (inappropriate varieties, lack of irrigation, land tenure issues), deepening the development gap. SSA was not part of the Green Revolution and suffers today from low productivity.

Beyond the Green Revolution: Current Food System Challenges

The Green Revolution raised food production but created a fragile, fossil-fuel-dependent, ecologically damaging food system. Current challenges include:

- **Climate change** — every 1°C of warming reduces wheat, maize, and rice yields by ~5–10%. Extreme weather events (droughts, floods, heatwaves) are disrupting production. India banned wheat exports in 2022 after a heatwave destroyed harvests; Pakistan's 2022 floods submerged 45% of its farmland.
- **Soil degradation** — 24 billion tonnes of fertile soil lost annually. At current rates, the world has ~60 harvests remaining in some intensively farmed soils. Regenerative agriculture (composting, cover crops, minimal tillage, crop rotation) is gaining attention as a sustainable alternative.
- **Water** — agriculture uses 70% of global freshwater; aquifer depletion is accelerating; climate change is altering rainfall patterns.
- **Biodiversity loss** — 75% of the world's food supply comes from just 12 plant species and 5 animal species. This extreme dependence on a narrow genetic base creates catastrophic vulnerability to new

pathogens (e.g. Fusarium wilt is threatening the Cavendish banana, the world's most traded fruit, just as Panama disease destroyed the Gros Michel banana in the 1950s).

- **Food systems and climate** — the global food system generates ~26% of global greenhouse gas emissions (Oxford University). Meat production alone accounts for ~14.5% of emissions. Dietary shift towards plant-based foods is identified as the single most impactful individual action for reducing food-related emissions (a vegan diet produces ~73% fewer GHGs than a high-meat diet).

11. Migration Case Study: The Syrian Refugee Crisis (Expanded)

Case Study: Syrian Refugee Crisis (2011–present) — Causes, Impacts, and Geopolitical Dimensions

Background: The Syrian Civil War began in March 2011 as part of the Arab Spring uprisings, when government crackdown on pro-democracy protests in Deraa escalated into armed conflict. The war became a multi-sided conflict involving: Assad's government (backed by Russia and Iran), various rebel groups (Western-backed), ISIS/ISIL (controlling large territories 2013–2018), Kurdish forces (YPG/SDF, backed by USA), and Turkish military interventions. The conflict created the largest displacement crisis since World War II.

Scale (2024):

- 6.5 million Syrian refugees registered with UNHCR in neighbouring countries and Europe
- 7.2 million internally displaced persons (IDPs) within Syria
- 90% of Syria's population lives below the poverty line
- 50%+ of Syria's infrastructure destroyed, including 40% of healthcare facilities
- A generation of children (now teenagers) with significant educational gaps

Impacts on host countries:

Turkey (3.3 million registered refugees — world's largest refugee-hosting country): estimated \$40+ billion total costs since 2011; majority housed outside camps in informal urban settlements; Syrians represent 4% of Turkey's population; public resentment rose during economic crisis (2021–22), making Syrian refugees a political lightning rod (President Erdogan promised "voluntary returns" during 2023 elections). Labour market impact: Syrians often work informally in agriculture, construction, and textile sectors, suppressing wages in low-skill sectors (disputed: some studies find net positive impact through increased consumer demand).

Lebanon (800,000–1.5 million, exact number disputed — ~25% of Lebanon's pre-crisis population):

Lebanon's economic collapse (2019–) has been exacerbated by refugee costs; schools operate double shifts (Lebanese children in morning, Syrian refugees in afternoon); water, electricity, and sanitation systems overwhelmed; deepening sectarian tensions (Lebanese political system based on delicate religious balance, threatened by influx of predominantly Sunni Muslim refugees). Some Lebanese politicians call for "safe return" even before security conditions are adequate.

Jordan (670,000 registered refugees, estimated 1.3 million including unregistered): Zaatari Camp (~150,000 residents at its 2013 peak, ~80,000 today) became Jordan's fourth largest city with a permanent market street. Jordan's water resources (among the world's most scarce) severely strained. However, international aid (billions in EU and US funding) partially offsets fiscal burden.

Germany (~1 million Syrians, largest population outside the Middle East): Angela Merkel's "Willkommenskultur" (welcome culture) in 2015 admitted 890,000 asylum seekers (various nationalities) in a single year. Strong humanitarian response but stretched asylum processing (decisions took 1–3 years). Political backlash: AfD (far-right) rose from 4.7% (2013) to 12.6% (2017) to 10.3% (2021) and 20.8% (2025 Federal Election). Integration outcomes: a large and rising share of Syrians in work; German language acquisition programmes; children integrated into schools. Example of successful humanitarian response despite political costs.

Root causes and structural factors:

The Syrian crisis illustrates how political causes interact with structural factors. Syria experienced years of extreme drought (2006–2010) driven partly by climate change, which drove 1.5 million rural Syrians to cities, strained urban resources and social services, and created the conditions for political discontent. Climate change is increasingly identified as a "threat multiplier" for conflict and displacement: it doesn't directly cause wars but exacerbates existing social, economic, and political tensions. Kelley et al. (2015) argued that climate change made a drought this severe 2–3 times more likely, although the strength of the drought–war link is debated.

Geopolitical dimensions: The crisis exposed deep divisions in the international refugee system. The 1951 Refugee Convention defines a refugee as someone persecuted for race, religion, nationality, political opinion, or membership of a social group. It does not cover environmental or climate refugees (a growing category). European states used the "safe third country" principle, Dublin Regulation, and offshore processing to limit legal pathways, pushing refugees towards dangerous irregular migration routes. The Central Mediterranean sea crossing has claimed 20,000+ lives since 2014. Turkey's 2016 "deal" with the EU (€6 billion in return for stemming migration flows) illustrates

how wealthy states use financial leverage to outsource refugee protection to poorer neighbouring countries.

Exam Tip: When discussing migration case studies in AQA exams, always cover: (1) causes of migration (push-pull, Lee's model), (2) nature of the flow (who migrates, how, via what routes), (3) impacts on source country/area, (4) impacts on destination country/area (both positive and negative), and (5) policy responses. The strongest answers evaluate the extent to which migration is "beneficial" or "problematic" — recognising that the answer depends on whose perspective you take (host government? local community? the migrant themselves? the source country?).

Exam Tip: The connection between population, environment, and migration is a key synoptic theme. Climate change is projected to displace 200 million–1 billion people by 2050 (various estimates). The Paris Agreement aims to limit warming to 1.5–2°C to reduce this displacement risk. This links Population and the Environment to Carbon and Water Cycles (Core), Resource Security (Option), and Changing Places (Option). Demonstrating these connections in essays earns synoptic marks.